

$$x^2 + 0^2 = 25$$

$$x^2 = 25$$

$$x = \pm 5$$

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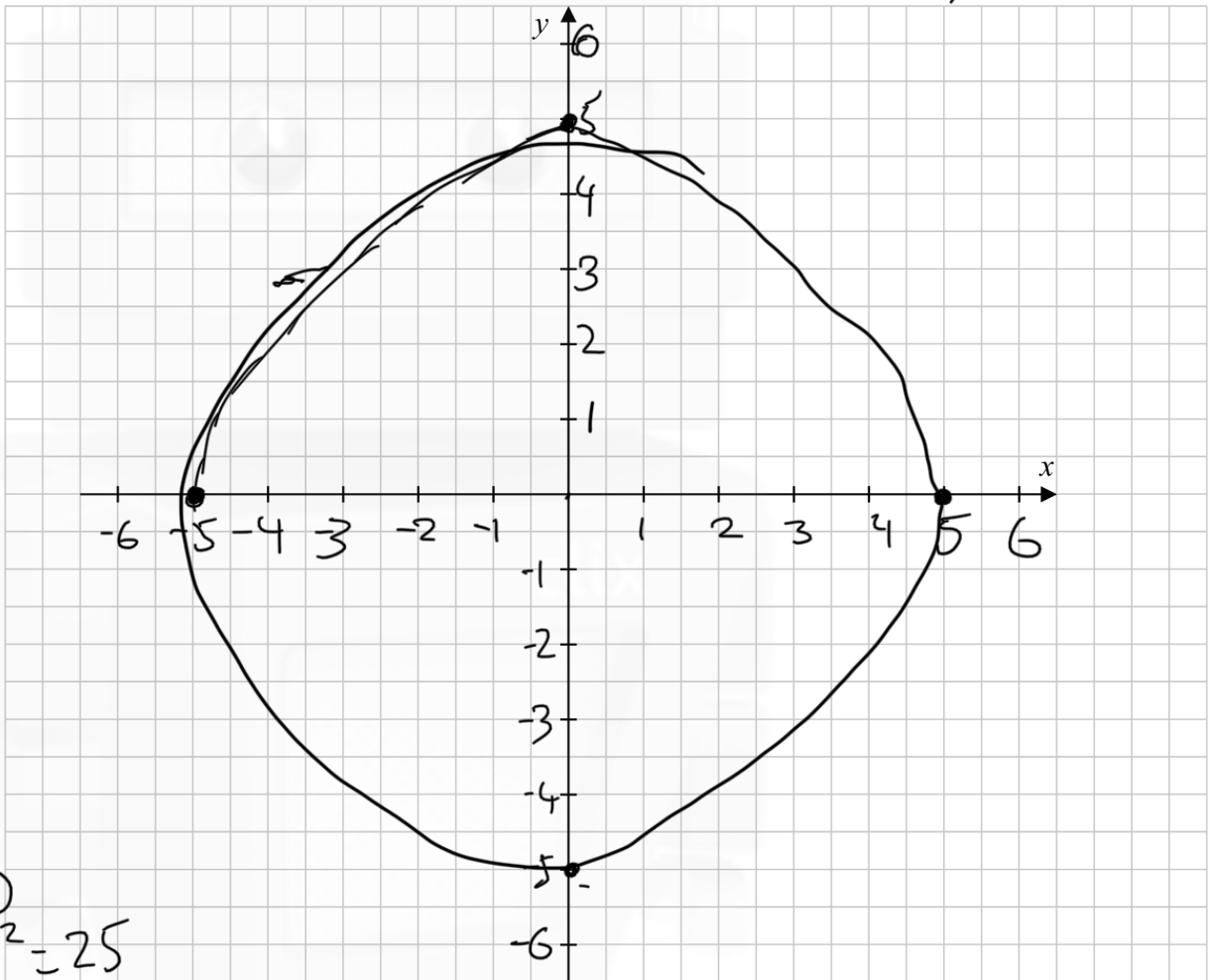
$$(5, 0)$$

$$(-5, 0)$$

(25 marks)

Question 3

(a) Draw the circle $c: x^2 + y^2 = 25$. Show your scale on both axes.



@ x = 0

$$0^2 + y^2 = 25$$

$$y = \pm 5$$

$$(0, 5)$$

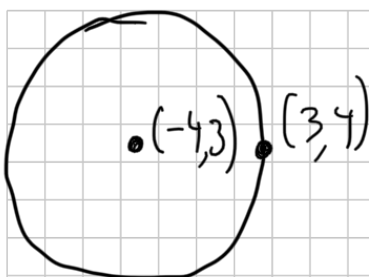
$$(0, -5)$$

(b) Verify, using algebra, that $A(-4, 3)$ is on c .

$$(-4)^2 + 3^2 = 25$$

$$16 + 9 = 25$$

(c) Find the equation of the circle with centre $(-4, 3)$ that passes through the point $(3, 4)$.



$$\sqrt{(3 - (-4))^2 + (4 - 3)^2}$$

$$\sqrt{(7)^2 + 1^2}$$

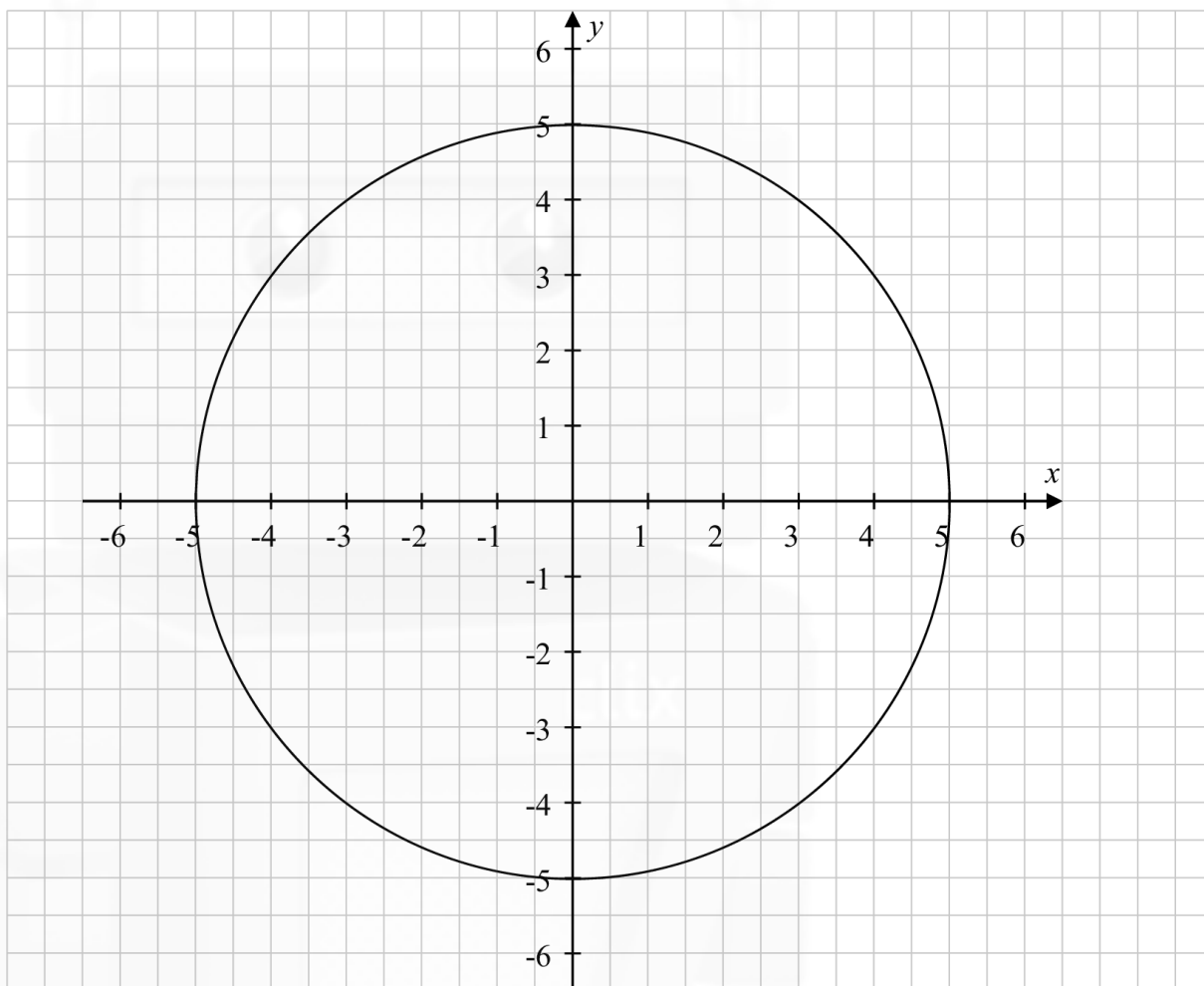
$$\sqrt{49 + 1}$$

$$\sqrt{50}$$

$$(x - (-4))^2 + (y - 3)^2 = \sqrt{50}^2$$

(x

- (a) Draw the circle $c: x^2 + y^2 = 25$. Show your scale on both axes.



- (b) Verify, using algebra, that $A(-4, 3)$ is on c .

$$x^2 + y^2 = 25$$

$$(-4)^2 + 3^2 = 16 + 9 = 25 = \text{RHS}$$

or

Centre of c : $O(0, 0)$

$$|OA| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(-4 - 0)^2 + (3 - 0)^2}$$

$$= \sqrt{25} = 5 = \text{radius of } c$$

- (c) Find the equation of the circle with centre $(-4, 3)$ that passes through the point $(3, 4)$.

$$r = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(3 + 4)^2 + (4 - 3)^2} = \sqrt{49 + 1} = \sqrt{50}$$

$$(x - h)^2 + (y - k)^2 = r^2 \Rightarrow (x + 4)^2 + (y - 3)^2 = (\sqrt{50})^2 = 50$$

- (a) (i) The circle c has equation $(x+2)^2 + (y-3)^2 = 100$.

Write down the co-ordinates of A , the centre of c .

$A(-2, 3)$

Write down r , the length of the radius of c .

$r = 10$

- (ii) Show that the point $P(-8, 11)$ is on the circle c .

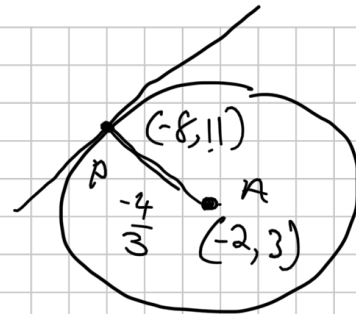
$$\begin{aligned} (-8+2)^2 + (11-3)^2 &= 100 \\ (-6)^2 + (8)^2 &= 100 \\ 36 + 64 &= 100 \end{aligned}$$

- (b) (i) Find the slope of the radius $[AP]$.

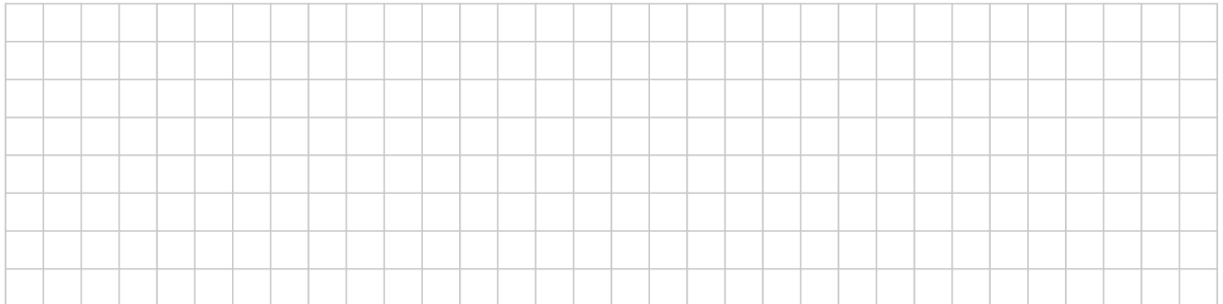
$$m = \frac{y_2 - y_1}{x_2 - x_1} \Rightarrow \frac{3 - 11}{-2 - (-8)} = \frac{-8}{6} = -\frac{4}{3}$$

- (ii) Hence, find the equation of t , the tangent to c at P .

$$\begin{aligned} y - 11 &= \frac{3}{4}(x - (-8)) \\ y - 11 &= \frac{3}{4}x - 6 \\ 4y - 44 &= 3x - 24 \\ 4y - 3x &= -24 + 44 \\ 3x - 4y &= 68 \end{aligned}$$



- (c) A second line k is a tangent to c at the point Q and $k \parallel t$. Find the co-ordinates of Q .



- (a) (i) The circle c has equation $(x+2)^2 + (y-3)^2 = 100$.

Write down the co-ordinates of A , the centre of c .

$$A(-2, 3)$$

Write down r , the length of the radius of c .

$$r = 10$$

- (ii) Show that the point $P(-8, 11)$ is on the circle c .

$$(-8+2)^2 + (11-3)^2 = 100$$

$$36 + 64 = 100$$

$$100 = 100$$

or

$$|AP| = \sqrt{(-8+2)^2 + (11-3)^2}$$

$$|AP| = \sqrt{100} = 10$$

- (b) (i) Find the slope of the radius $[AP]$.

$$\text{Slope of radius} = -\frac{4}{3}$$

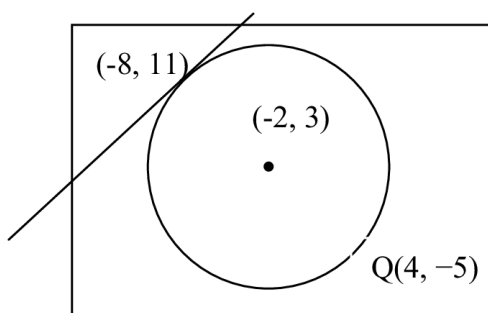
- (ii) Hence, find the equation of t , the tangent to c at P .

$$\perp = \frac{3}{4}$$

$$y - 11 = \frac{3}{4}(x + 8)$$

$$3x - 4y + 68 = 0$$

- (c) A second line k is a tangent to c at the point Q and $k \parallel t$. Find the co-ordinates of Q .



$$(-8, 11) \rightarrow (-2, 3) \quad x \uparrow 6, y \downarrow 8 \rightarrow Q(4, -5)$$

